Kiran Tomlinson

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Employment

Senior Researcher 2024–
Microsoft Research, AI Interaction and Learning Group

Visiting Instructor 2023
Carleton College, Department of Computer Science

Education

PhD, Computer Science 2019–2024
Cornell University
Advisor: Jon Kleinberg

MS, Computer Science 2019–2022
Cornell University

BA, Computer Science and Mathematics 2015–2019
Carleton College
Summa Cum Laude, Distinction in Computer Science and in Mathematics

Internships

Microsoft Research Summer 2022
Productivity and Intelligence Group, with Jennifer Neville

Microsoft Summer 2021
Office of Applied Research, with Longqi Yang and Mengting Wan

NASA Johnson Space Center Summers 2016 & 2017
Spacecraft Software Engineering Branch, with Nathan Uitenbroek

Publications

*Equal contribution, (α–β) alphabetical order

21. Working with AI: Measuring the Applicability of Generative AI to Occupations
Kiran Tomlinson, Sonia Jaffe, Will Wang, Scott Counts, and Siddharth Suri
NeurIPS, 2026

20. Adopt ≠ Adapt: Longitudinal Analyses of LLM Conversations in the Wild
Rebecca M. M. Hicke and **Kiran Tomlinson**
EMNLP, 2026

19. AI and the Democratization of Knowledge Work
Madeleine I. G. Daepp, **Kiran Tomlinson**, Scott Counts, Siddharth Suri
Nature Computational Science, 2026

18. Reasoning about Reasoning: BAPO Bounds on Chain-of-Thought Token Complexity in LLMs
Kiran Tomlinson, Tobias Schnabel, Adith Swaminathan, and Jennifer Neville
ICML, 2026

17. Exclusion Zones of Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
AAAI, 2026

16. Lost in Transmission: When and Why LLMs Fail to Reason Globally
Tobias Schnabel*, **Kiran Tomlinson***, Adith Swaminathan, and Jennifer Neville
NeurIPS (spotlight, 3%), 2025

15. When the Universe is Too Big: Bounding Consideration Probabilities for Plackett–Luce Rankings
Ben Aoki-Sherwood, Catherine Bregou, David Liben-Nowell, **Kiran Tomlinson**, and Thomas Zeng (α–β)
AISTATS, 2025

14. Replicating Electoral Success
Kiran Tomlinson, Tanvi Namjoshi, Johan Ugander, and Jon Kleinberg
AAAI, 2025

13. Computational Perspectives on Individual and Collective Decision-Making
Kiran Tomlinson
PhD Dissertation, 2024

12. The Moderating Effect of Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
AAAI, 2024

11. Graph-Based Methods for Discrete Choice
Kiran Tomlinson and Austin R. Benson
Network Science 12(1), 2023

10. Workplace Recommendation with Temporal Network Objectives
Kiran Tomlinson, Jennifer Neville, Longqi Yang, Mengting Wan, and Cao Lu
KDD, 2023

9. Targeted Training for Multi-organization Recommendation
Kiran Tomlinson, Mengting Wan, Cao Lu, Brent Hecht, Jaime Teevan, and Longqi Yang
ACM Transactions on Recommender Systems 1(3), 2023

8. Ballot Length in Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
AAAI, 2023

7. Choice Set Confounding in Discrete Choice
Kiran Tomlinson, Johan Ugander, and Austin R. Benson
KDD, 2021

6. Learning Interpretable Feature Context Effects in Discrete Choice
Kiran Tomlinson and Austin R. Benson
KDD, 2021

5. Choice Set Optimization Under Discrete Choice Models of Group Decisions
Kiran Tomlinson and Austin R. Benson
ICML, 2020

4. Summarizing Diverging String Sequences, with Applications to Chain-Letter Petitions
Patty Commins, David Liben-Nowell, Tina Liu, and **Kiran Tomlinson** (α–β)
31st Annual Symposium on Combinatorial Pattern Matching (CPM), 2020

3. Distance Measures for Tumor Evolutionary Trees
Zach DiNardo*, **Kiran Tomlinson***, Anna Ritz, and Layla Oesper
Bioinformatics 38(7), 2020

2. Parameter, Noise, and Tree Topology Effects in Tumor Phylogeny Inference
Kiran Tomlinson and Layla Oesper
BMC Medical Genomics 12(10), 2019

1. Examining Tumor Phylogeny Inference in Noisy Sequencing Data
Kiran Tomlinson and Layla Oesper
IEEE Int. Conf. on Bioinformatics and Biomedicine (BIBM), 2018

Preprints

3. Putting Transformer Theories to the Test
Awni Altabaa, Tobias Schnabel, **Kiran Tomlinson**
arXiv, 2026

2. Code Understanding is a Bottleneck for Coding Agents
Nishant Balepur, **Kiran Tomlinson**, Tobias Schnabel
arXiv, 2026

1. AI in the Enterprise: How People Use M365 Copilot Chat
   Scott Counts, Yan Chen, Jing Dong, Himanshu Sharma, Andrey Zaikin, Rui Hu, Alperen Kok, Gorkem Ozer Yilmaz, Siddharth Suri, **Kiran Tomlinson**, Sonia Jaffe, Will Wang
   arXiv, 2026

Patents

2. Intelligent system and method of optimizing cross-team information flow
   Longqi Yang, Mengting Wan, Cao Lu, Jennifer Neville, and **Kiran Tomlinson**
   US12586038B2, Mar 24, 2026 (filed Apr 20, 2023)

1. Targeted training of inductive multi-organization recommendation models for enterprise applications
   Kiran Tomlinson, Longqi Yang, Mengting Wan, Cao Lu, Brent Hecht, and Jaime Teevan
   US11983649B2, May 14, 2024 (filed Oct 26, 2021)

Invited Talks

"BAPO: A Communication Complexity Model of Transformer Capabtilities." UW Theory Seminar, Oct 2026.

"Working with AI: Measuring the Applicability of Generative AI to Occupations." IEEE AIIoT '26 (Keynote), May 2025.

"Exclusion Zones of Instant Runoff Voting." COMSOC Video Seminar, Feb 2026.

"Working with AI: Measuring the Applicability of Generative AI to Occupations." Louisiana State Board of Elementary and Secondary Education, Sep 2025.

"Algorithmic Perspectives on Instant Runoff Voting." Cornell CS Theory Seminar, Aug 2023.

"Choices in Networks." NetSci–SINM, Jul 2023.

"A Computational Look at Ranked-Choice Voting." Carleton College CS Tea, May 2023.

"Choice Models for Networks and Networks for Choice Models." Network Dynamics and Choice Theory Workshop @ UVM, May 2022.

"Feature Context Effects and Choice Set Confounding in Discrete Choice." WPI, CS 525, Apr 2021.

"Feature Context Effects and Choice Set Confounding in Discrete Choice." Cornell/MIT MURI, Mar 2021.

"Learning Interpretable Feature Context Effects in Discrete Choice." Stanford Ugander Lab, Dec 2020.

Workshop Talks

"Working with AI: Measuring the Applicability of Generative AI to Occupations." NBER AI and Economic Measurement Workshop, May 2026.

"Working with AI: Measuring the Applicability of Generative AI to Occupations." NABE Tech Economics Conference (Plenary), Nov 2025.

"Working with AI: Measuring the Applicability of Generative AI to Occupations." Wharton Business & Generative AI Conference, Sep 2025.

"Exclusion Zones in Instant Runoff Elections." AMS Special Session on Mathematics of Decisions, Elections, and Games @ JMM, Jan 2025.

"Workplace Recommendation with Temporal Network Objectives." NetSci, Jul 2023.

"Learning Context Effects in Triadic Closure." NetSci–SINM, Sep 2020.

"Distance Measures for Tumor Evolutionary Trees." RECOMB–CCB (with Zach DiNardo), May 2019.

Awards

The Web Conference Outstanding Reviewer 2024
Phi Beta Kappa 2019
Sigma Xi 2019
CRA Outstanding Undergraduate Researcher Honorable Mention 2018

Goldwater Scholarship Honorable Mention 2018
Patricia V. Damon Scholarship 2018
NSF Student Travel Award (to BIBM) 2018
Carleton College Dean's List 2016, 2017, 2018
National Merit Scholarship 2015–2018

Teaching

Carleton College (Instructor)
CS 201: Data Structures — Winter 2023 (×2), Spring 2023
CS 202: Mathematics of Computer Science — Spring 2023

Cornell University (Teaching Assistant)
CS 2110: Object-Oriented Programming and Data Structures — Fall 2019
CS 2800: Discrete Structures — Spring 2020

Carleton College (Prefect)
CS 201: Data Structures — Winter 2018, Spring 2019
CS 202: Mathematics of Computer Science — Fall 2017, Spring 2018
CS 254: Computability and Complexity — Fall 2018, Winter 2019

Service

Workshop Program Chair
What Can('t) Transformers Do? @ NeurIPS '25
The Workshop on Human and Machine Decisions @ NeurIPS '21

Program Committee
AAAI '23, '24, '25, '26, '27
AISTATS '25, '26, '27
ACL ARR '26
ICLR '26, '27
ICML '25, '26
ICWSM '25
NeurIPS '24, '25, '26
WebConf '24, '25, '26

Journal Reviewer
Science Advances
Transactions on Machine Learning Research
Biometrika
Data Mining and Knowledge Discovery
Annals of Mathematics and Artificial Intelligence
Information Processing and Management

PhD Admissions Committee 2021, 2022
Cornell University Computer Science Department

Graduate and Professional Student Assembly Representative (elected) 2019–2020
Cornell Computer Science Graduate Organization

Certifications

Private Pilot (ASEL), FAA 2026
Amateur Radio Operator (Technician), FCC 2017
Open Water Diver, PADI 2015